Experiment 2: Train-Test Split and K-Fold Cross-Validation Using KNN

Aim:

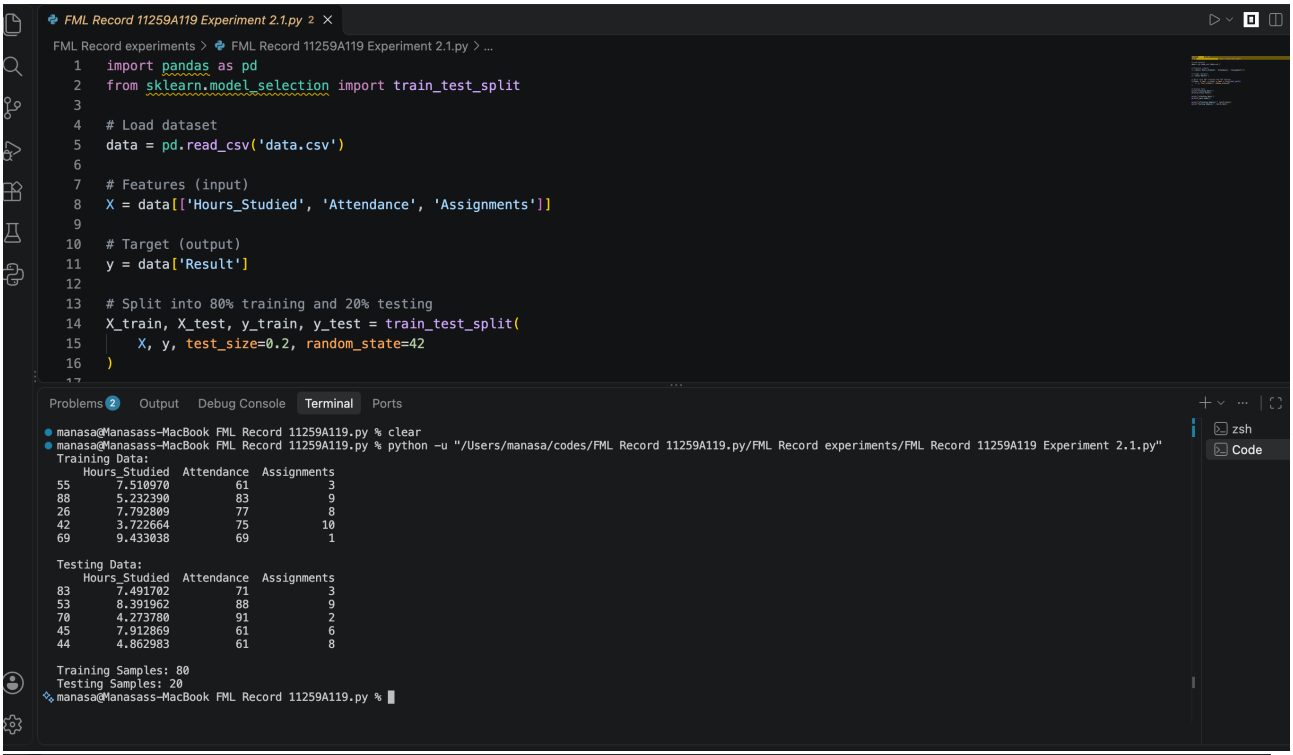
To split the dataset into training and testing sets and evaluate the performance of a K-Nearest Neighbors (KNN) classifier using 5-fold cross-validation.

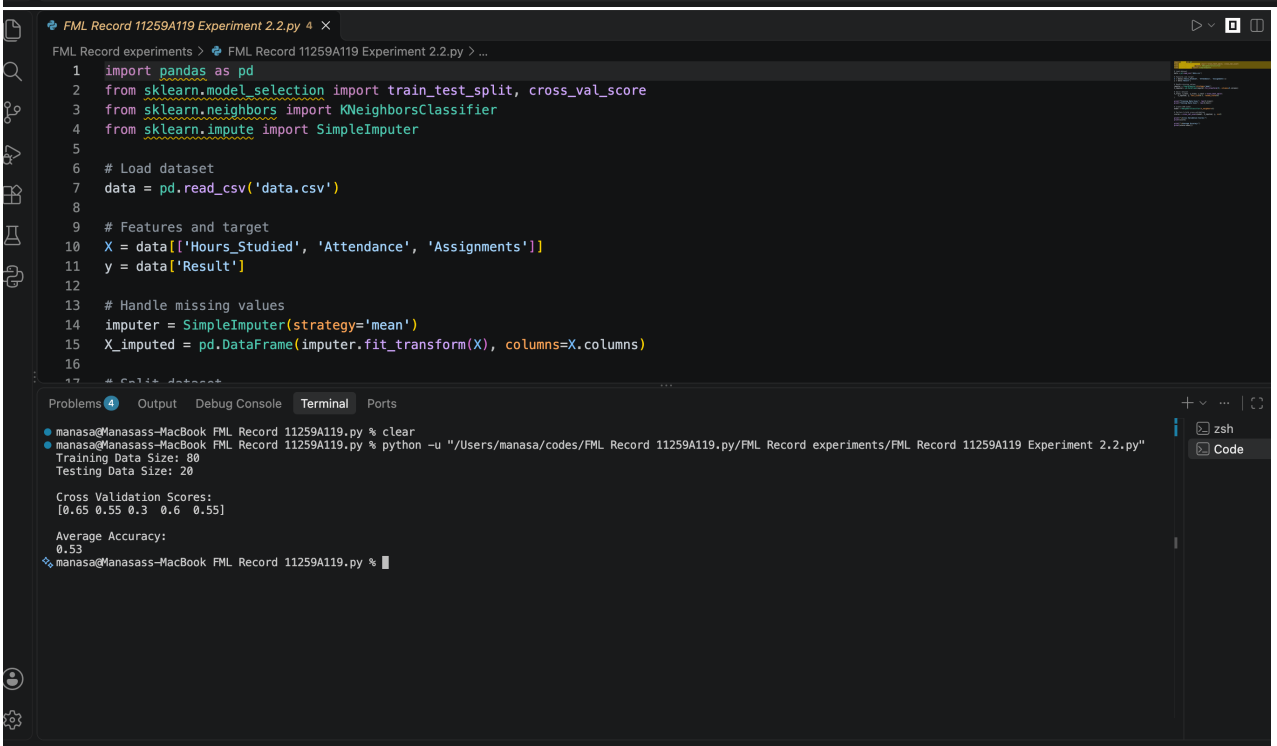
Algorithm:

1. Import the required libraries.
2. Load the dataset and separate the input features and target variable.
3. Split the dataset into training and testing sets using `train_test_split()`.
4. Create a KNN classifier.
5. Apply 5-fold cross-validation using `cross_val_score()`.

Calculate the average cross-validation accuracy.

7. Train the classifier on the training data and evaluate its performance on the test data.





Result:
Thus, the dataset was divided into training and testing sets, and the KNN classifier was evaluated using 5-fold cross-validation to determine its average accuracy.